

Review article

Green artificial intelligence and its lifecycle, hardware, and measurement dimensions

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ABSTRACT

Context: Across the artificial intelligence (AI) lifecycle — from hardware provisioning and development to deployment, operation, and end-of-life — environmental burdens span energy, CO₂e emissions, water use, and embodied/material footprints. Cloud-provider tools can improve operational visibility but remain heterogeneous and only partially cover water and value-chain effects, limiting comparability and reproducibility.

Objective: Addressing these multidimensional burdens benefits from a lifecycle-oriented perspective linking phase-explicit structuring with system levers such as hardware context, placement, energy mix, cooling, and scheduling, together with calibrated measurement across facility, system, device, and workload levels.

Methods: This systematic review synthesizes 103 studies and, within the lifecycle-focused analysis, extracts and harmonizes 164 Green AI-related terms. Included studies were coded along three dimensions: lifecycle phase, environmental impact category, and system level. The identified concepts were analytically related to Life Cycle Assessment (LCA) stages to structure the lifecycle perspective. Measurement approaches were compared by distinguishing estimator-based reporting from metered or telemetry-based reference data, including calibration procedures and reporting completeness.

Results: The review derives a synthesized definition of Green AI that provides a conceptual reference point for the lifecycle, hardware, and measurement analyses. It synthesizes a five-phase lifecycle structure and analytically relates it to the four established LCA stages. It further synthesizes a class-based hardware perspective across the edge-cloud continuum and a hybrid measurement logic that relates estimator-based approaches to metered or telemetry-based reference data to support calibration and more consistent reporting. Important gaps remain, including inconsistent indicators, limited treatment of water and Scope 3 impacts, provider-specific dashboards that hinder comparability, and the lack of reproducible, calibrated measurements across hardware tiers.

Conclusion: By integrating definition, lifecycle structuring, hardware context, and calibrated measurement, this article offers an analytically grounded synthesis of Green AI relevant to researchers and practitioners. It supports more structured discussion of green model and infrastructure choices, context-aware scheduling, transparent reporting, and circularity-oriented future work.

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1. Introduction

AI accelerators, especially GPUs and domain-specific tensor or neural processing units, have become central compute platforms for training transformer-based and large-scale multimodal models [1,2]. Realizing these capabilities has entailed a substantial scale-up in model size and training compute, from millions to billions of parameters in publicly reported models, together with operations such as matrix multiplications and attention mechanisms [3,4]. As a result, training large deep neural networks and large language models can require substantially higher energy demand than many conventional machine-learning approaches, depending on model size, training duration, and infrastructure [5,6]. Beyond operational energy and carbon emissions, recent approaches emphasize additional environmental burdens — including water consumption and embodied environmental impacts across the hardware supply chain — that remain underrepresented in model-centric accounts [7–10]. These environmental burdens have motivated the emerging research field referred to as Green AI.

At the same time, cloud providers increasingly expose emissions tooling, such as dashboards for location- and market-based accounting, which improves operational transparency. Methodologies, however, differ across providers and only partially cover water and Scope 3 value-chain effects, which limits cross-system comparability [11,12]. At the application level, many contemporary AI systems combine multiple model and service components across development and deployment pipelines. They can increase energy demand at deployment when subjected to large request volumes, long context windows, strict latency service-level objectives, and heterogeneous edge–cloud placement constraints [13].

Despite notable progress, the field lacks widely adopted AI-specific reporting practices for lifecycle-wide and calibrated assessment that integrates energy, CO₂e emissions, water use, and embodied/material footprints and enables reproducible comparison across hardware tiers and deployment contexts. Existing work often addresses specific segments, such as data centers, algorithms, or runtime kernels. The connections between these segments across the AI lifecycle remain less consistently developed [13,14].

Accurate assessment of Green AI depends on coordinated, multi-level measurement. This may include facility-level indicators such as Power Usage Effectiveness (PUE) and Water Usage Effectiveness (WUE), system-level consumption, hardware telemetry such as RAPL and PMCs, workload profiling, and lifecycle-related coverage of hardware, development, deployment, and end-of-life [13,15]. Where metered or telemetry-based reference data are available, indirect estimators should be checked against them. This makes reported values easier to interpret and compare.

The review is guided by the following research questions:

- **RQ1:** How should *Green AI* be defined and delimited relative to adjacent concepts, and which recurring principles are discussed in the literature in relation to its lifecycle-wide application?
- **RQ2:** Which phases and subphases structure the *Green AI lifecycle*, and how can they be analytically related to the stages of Life Cycle Assessment (LCA)?
- **RQ3:** Which hardware components and system architectures are associated with improved performance-per-watt and potentially reduced embodied environmental impacts across the lifecycle?
- **RQ4:** How can environmental impacts (e.g., energy, CO₂e emissions, water use, and embodied/material footprints) be *measured and reported* more consistently across the lifecycle, including calibration of indirect estimators with metered or telemetry-based reference data?
- **RQ5:** Which open *challenges and research priorities*, such as benchmarks, reporting schemas, uncertainty handling, circularity, and governance-related conditions, are needed to advance lifecycle integration of Green AI?

Contributions. This article advances the literature through four key contributions: (1) a synthesized conceptual definition of Green AI and a proposed review-based analytical completion definition of Green AI, (2) a lifecycle-oriented analytical structure, (3) a class-based hardware perspective, and (4) a synthesized measurement logic. Building on these contributions, the article further derives analytically grounded directions for future Green AI research.

The remainder of the article is structured as follows. Section 2 surveys the core Green AI literature and selected adjacent perspectives from economics and sustainability research. Section 3 details the methodology, including taxonomy-driven coding and trend analysis. Section 4 reports findings addressing RQ1–RQ4. Section 5 synthesizes major challenges and outlines the research agenda (RQ5). Section 6 discusses implications for model and infrastructure selection, carbon- and context-aware scheduling, lifecycle-oriented evaluation, and iterative improvement. It also addresses limitations and threats to validity, including SLR coverage bias, estimator-meter discrepancies, provider-specific dashboards, and incomplete treatment of water and Scope 3 impacts. Section 7 concludes the article by summarizing the main findings and future research directions.

2. Related work

Green artificial intelligence (AI) research has increasingly addressed the tension between maintaining task performance and reducing environmental impacts, including operational energy use and carbon emissions, cooling-related water use, and material or embodied footprints

in hardware supply chains [7,8,14]. Early evidence on the training-related footprint of large models motivated subsequent studies along complementary levers, including algorithmic efficiency, hardware and placement choices, compiler/runtime optimizations, and data and storage management, often with attention to energy or carbon reduction under quality or latency constraints [5]. Within the algorithmic family, pruning and quantization, together with related compression and adaptation methods (e.g., knowledge distillation, low-rank adaptation, and sparsity), can reduce computational demand and may lower energy use when implemented under suitable hardware and workload conditions, while aiming to control accuracy loss [16].

Operationalizing Green AI. Operationalization spans multiple system layers. On the hardware side, performance-per-watt efficient AI accelerators affect computational efficiency, while context-aware placement across the edge–cloud continuum affects carbon and water-related impacts through regional energy mixes and cooling technologies, including cooling-water intensity [10,17]. At the runtime level, dynamic voltage and frequency scaling, power capping, queue-level energy budgeting, and locality-aware scheduling provide established High-Performance Computing (HPC) practices that can inform energy-aware AI pipelines. Data and storage practices also matter. De-duplication, tiered storage, retention policies, and data-quality controls can limit unnecessary data movement and redundant computation [18,19]. Green software engineering frameworks, including GAISSA [20], articulate concrete development and inference practices that are relevant to AI-centric systems while also highlighting additional hardware- and model-induced impacts specific to AI deployments.

Lifecycle-oriented and architecture-level perspectives. Green software models propose quality attributes and indicators — such as energy consumption, resource usage, and system longevity — that can inform architecture-level assessments of AI-based systems [18,19]. Complementary studies argue for end-to-end integration of energy-aware practices across lifecycle phases [21] and for relating AI processes to standardized LCA concepts, which can help make upstream effects (raw-material extraction, component manufacturing, transport) and downstream effects (inference, maintenance, reuse) visible alongside model-centric impact indicators.

Life Cycle Assessment as an analytical perspective for AI systems. LCA provides a standardized methodology for evaluating environmental impacts across the complete lifecycle of products and technical systems. LCA distinguishes four methodological stages: goal and scope definition, life cycle inventory, life cycle impact assessment, and interpretation. In sustainability research on digital technologies, LCA is increasingly used to assess environmental impacts beyond operational energy consumption, including embodied environmental impacts from hardware manufacturing, cooling infrastructures, and material extraction [7,8,22,23]. However, existing Green AI research frequently concentrates on training efficiency and computational optimization, while broader lifecycle perspectives remain comparatively underrepresented [5,9,14]. To address this gap, this article uses a review-derived analytical lifecycle structure to relate AI development and deployment activities to established LCA stages without implying a full quantitative LCA. This analytical relation provides a structured perspective for discussing environmental impact categories across the AI system lifecycle, including hardware provisioning, model development, deployment infrastructures, operational inference, and system retirement.

Positioning. Prior studies deliver important building blocks, but the reviewed literature remains fragmented across algorithmic, hardware, software-practice, and socio-economic perspectives. Adjacent platform literature further illustrates that conceptual delimitation becomes important when socio-technical infrastructure and ecosystem boundaries are analyzed across complex deployment contexts [24,25]. An integrated perspective that jointly connects lifecycle structuring, hardware context, and measurement remains limited in the literature. Based on the structured analysis of the reviewed literature, the paper presents a review-derived analytical synthesis of recurring elements rather than proposing a new prescriptive framework. The synthesis distinguishes “Green AI” from broader “Sustainable AI” perspectives and analytically relates lifecycle phases and subphases to LCA stages.

3. Methodological approach to this research

This study follows a systematic literature review (SLR) design and reports the review process in line with established guidance for evidence synthesis in information systems and software engineering, complemented by PRISMA 2020 reporting principles [26–29]. The overall review design was further structured using the literature review taxonomy perspectives of vom Brocke et al. [30] and Cooper [31], supporting a conceptually organized and lifecycle-oriented synthesis of Green AI research (cf. Online Appendix, Figure A.1). The literature search covered Web of Science, ScienceDirect, and IEEE Xplore as primary databases (cf. Online Appendix, Figure A.2). To improve coverage of leading AI venues that are not always consistently indexed in these databases, arXiv was additionally used for targeted retrieval and citation chaining. Preprints were explicitly flagged during screening and replaced by peer-reviewed versions where available; otherwise, they were used only for mapping or contextualization unless they introduced distinct methodological assets. The search strings were iteratively refined along three concept groups: (i) domain and context, (ii) environmental impact dimensions, and (iii) AI scope. This structure broadened the search beyond model-centric efficiency studies to include contributions on lifecycle boundaries, hardware, infrastructure, and environmental measurement.

The review process resulted in a final corpus of 103 studies after systematic identification ($n = 1695$), de-duplication and filtering, title/abstract screening ($n = 757$), full-text eligibility assessment, and backward/forward citation chaining. After title/abstract screening, 98 full-text publications were assessed for eligibility. At the full-text stage, 2 publications were excluded because they did not meet the inclusion criteria, leaving 96 database-sourced studies. Backward and forward citation chaining contributed 7 additional eligible studies after de-duplication and eligibility checks. Eligibility primarily focused on peer-reviewed journal articles and conference proceedings. Selected preprints were retained only when no peer-reviewed version was available and when they introduced distinct methodological or empirical assets relevant to the analysis; such cases were explicitly flagged during screening. Non-English publications, gray literature, and publications before 2014 were excluded to improve terminological consistency and focus the review on the modern deep-learning era (cf. Online Appendix, Figure A.3).

In methodological terms, included studies had to address AI and meet at least one of the following conditions: (i) quantitative reporting of energy, power, carbon, water, or embodied/material impacts; (ii) lifecycle boundary specification or mapping to Life Cycle Assessment (LCA) stages; (iii) measurement, telemetry, or calibration procedures; or (iv) Green AI definitions or definition-related formulations relevant to conceptual delimitation. Review studies were used primarily for mapping, conceptual comparison, and citation chaining. They were not treated as primary evidence for empirical effects unless they introduced distinct methodological assets such as boundary definitions, calibration procedures, or measurement models.

To ensure analytical transparency, a structured extraction schema was applied. It captured bibliographic information, lifecycle phases and their linkage to LCA stages, environmental impact dimensions, measurement approaches, hardware and deployment context, algorithmic and system-level optimization levers, and reporting completeness and reproducibility characteristics. In addition, the extraction schema recorded methodological provenance, such as emission-factor assumptions, regional context, and measurement scope. To reduce subjective bias and to preserve traceability between reported lifecycle activities and environmental indicators, the coding scheme and ambiguous category assignments were reviewed jointly by both authors.

Beyond the descriptive synthesis of the 103-study corpus, the review also included a focused lifecycle-coverage analysis for studies that explicitly addressed lifecycle-related concepts through identifiable lifecycle terminology, activities, boundaries, or indicators. For this

purpose, a lifecycle-relevant subcorpus of 43 studies meeting these criteria was analyzed in greater depth.

The lifecycle phases and subphases were derived through a six-step analytical procedure: (i) extraction and harmonization of lifecycle-related terminology, (ii) quantitative term analysis, (iii) chronological heatmapping of salient terms, (iv) hierarchical clustering of term co-occurrence, (v) aggregation of terms into subphases and higher-level lifecycle phases, and (vi) phase-level coverage analysis using a binary source-criterion matrix. This procedure was used to derive review-based analytical categories for the lifecycle synthesis in a transparent and traceable way, rather than to define prescriptive process steps. In addition to synthesizing Green AI literature, the review applies a lifecycle-oriented analytical lens derived from Life Cycle Assessment (LCA). No quantitative LCA is carried out; LCA stages function as an analytical reference for structuring lifecycle-wide environmental impacts. After aggregation, the extracted elements were assigned to analytical lifecycle phases. This made it possible to examine impacts beyond operational energy use. Detailed descriptions of the review process and supporting materials are provided in the Online Appendix (cf. Online Appendix, Table A.1).

4. Literature analysis

The corpus was organized into four overlapping analytical streams in order of size. The *largest stream* comprises lifecycle-oriented contributions (46/103 studies, 44.7%) that define stages, map activities to LCA, specify system boundaries, or synthesize stage-specific levers; these are analyzed in Section 4.2. A *second stream* comprises definition-oriented contributions (33/103 studies, 32.0%), which are analyzed in Section 4.1. A *third stream* concentrates on measurement assets (21/103 studies, 20.4%), including metered and calibrated-estimator approaches, telemetry, and emission-factor provenance; these are analyzed in Section 4.4. A *smaller but growing stream* targets green hardware and infrastructure (13/103 studies, 12.6%), including embodied/material accounting, energy-proportional architectures, DVFS, RAPL-based power capping, and carbon-aware scheduling; these are analyzed in Section 4.3. Since a study may belong to multiple categories, the presented counts reflect multi-label coding (cf. Online Appendix, Figure B.4 and Figure C.5). The following analysis proceeds stream-wise and synthesizes the evidence on Green AI. These streams provide the building blocks for characterizing Green AI and its lifecycle, hardware, and measurement dimensions.

4.1. Definition of Green AI

This section addresses RQ1 — “How should *Green AI* be defined and delimited relative to adjacent concepts, and which recurring principles are discussed in the literature in relation to its lifecycle-wide application?” — by examining explicit Green AI definitions and definition-related formulations in the literature (cf. Online Appendix, Table C.2), identifying recurring analytical limitations relevant to delimitation and operationalization, and deriving a synthesized definition.

Current research and practice increasingly report non-trivial environmental burdens of AI systems, including energy use, greenhouse-gas emissions, water use, and embodied/material impacts across development, deployment, operation, and supporting infrastructure. The magnitudes depend strongly on hardware choice, data center region, scheduling, workload characteristics, and reporting practices [5,9,10,32]. This makes a precise and operationally usable definition of Green AI necessary.

The literature does not converge on one narrow and consistently delimited definition. Early accounts primarily frame Green AI in compute- or efficiency-centered terms, whereas more recent contributions increasingly broaden the concept toward wider environmental footprint,

lifecycle, or umbrella sustainability perspectives. However, this broadening has not been accompanied by a corresponding stabilization of conceptual boundaries or operational criteria.

Table C.2 shows that the literature varies systematically along three dimensions. First, *conceptual emphasis*: from narrow compute- or efficiency-centered formulations to broader environmental footprint framings. Second, *scope and boundaries*: from training- or operation-centered views to broader lifecycle-oriented framings that may also include deployment, infrastructure, and embodied or material impacts. Third, *degree of operationalization*: from high-level appeals or umbrella framings to more specific formulations, although sharply delimited system boundaries, indicators, and assessment criteria are rarely specified in the definition itself [9,10,14,33–37].

Against this background, the remainder of this subsection critically appraises these recurring analytical limitations and uses them as the basis for a synthesized definition.

4.1.1. Recurring shortcomings in existing Green AI definitions

The following paragraphs summarize the recurring shortcomings identified across existing definitions of Green AI.

System-level perspective. Existing definitions of Green AI remain predominantly centered on AI research outputs, model quality, or computational efficiency, rather than on the realized AI system as a whole [14,33,36]. Schwartz et al. define Green AI in terms of obtaining novel results while accounting for computational cost, and Xu et al. retain this efficiency-oriented focus by emphasizing lightweight and efficient technologies for training and inference. Tornede et al. move beyond individual learning algorithms by considering machine learning pipelines and the overall software solution. Their focus, however, remains tied to Green AutoML rather than to a general Green AI definition [14,33,36]. Environmental impacts depend on how models are implemented across software, data, hardware, and deployment contexts. Patterson et al. show that energy use and CO₂e depend on the processor, data center efficiency, and carbon intensity of the energy supply, not only on the model. Yokoyama et al. similarly identify software, hardware, and energy-source carbon intensity as direct determinants of machine-learning energy consumption, while Chen et al. note that larger AI models require more powerful platforms for deployment and real-time implementation [34,35,38]. Taken together, these findings indicate that Green AI cannot be assessed adequately at the level of algorithms or model architectures in isolation. A synthesized definition should therefore adopt a system-level perspective and treat AI as a realized system whose environmental properties emerge only in the interaction of models, software, hardware, infrastructure, and socio-economic deployment context.

Boundary-explicitness and referent delimitation. Dodge et al. formalize operational software carbon intensity through explicit reference to energy use, carbon intensity, embodied emissions, and a declared functional unit, thereby illustrating that environmental claims depend on the chosen accounting scope [17]. Complementing this, Kaack et al. distinguish between computing-related impacts, immediate application effects, and broader system-level effects, which further underlines the need to specify what exactly is included in a Green AI claim [32]. A synthesized definition should therefore make its referent explicit and require declared system and lifecycle boundaries for the environmental impacts under consideration.

Delimitation from adjacent umbrella concepts and broad framings. Existing Green AI definitions and related formulations are not always sharply delimited from broader umbrella concepts. In particular, some contributions frame Green AI in ways that partially overlap with AI for sustainability, broader sustainability-oriented AI initiatives, or socio-technical conceptions of Sustainable AI [9,10,37].

This is particularly visible in broad review-based formulations. Verdecchia et al. define Green AI in terms of “practices” that address either the environmental impact of AI itself or the use of AI

to mitigate human environmental impacts. Alzoubi and Mishra organize Green AI initiatives into several initiative types, including infrastructure optimization, model-efficiency approaches, footprinting tools, sustainability-oriented development practices, open-source activities, and community-building efforts. Their account therefore approaches Green AI from an “AI-technology-application” perspective rather than from a strict conceptual-definition perspective. Bolón-Canedo et al. likewise integrate both *green-in AI* and *green-by AI* “paradigms” into their discussion of Green AI, thereby combining environmentally efficient AI design with AI applications for environmental sustainability [9,10,37].

Lifecycle perspective. Existing definitions of Green AI often remain focused on isolated phases, especially computational cost during model development or training, while only later work begins to broaden the perspective toward deployment, operation, and wider lifecycle-related considerations [10,14,33,34]. Schwartz et al. and Xu et al. define Green AI primarily in terms of “AI research” obtaining novel or competitive results with lower computational cost, and Xu explicitly discusses energy usage and carbon emissions during training and inference rather than a lifecycle-explicit environmental accounting perspective [14,33]. Patterson et al. broaden the discussion by distinguishing operational from lifecycle emissions and by considering ML energy use across research, development, and production, even though their analysis remains focused on operational training emissions [34]. Bolón-Canedo et al. move further toward a broader framing by explicitly referring to model design, training, and deployment, and by combining green-in AI, green-by AI, measurement tools, and regulatory aspects under one Green AI umbrella [10].

Taken together, these contributions suggest that Green AI cannot be defined adequately at the level of training or inference alone. Environmental impacts arise across multiple stages, including data-related processes, model development, deployment, operation, and supporting infrastructure. Verdecchia et al. make this broader point explicitly by arguing that achieving Green AI requires joint effort across different stages of an AI system’s lifecycle, such as data collection, training, and monitoring [9]. A synthesized definition should therefore adopt a lifecycle perspective so that cumulative environmental effects can be attributed more consistently across the relevant stages, rather than being reduced to isolated efficiency gains in a single phase.

AI-related scenario-based evaluation. Existing definitions of Green AI usually remain at a high level of abstraction and do not distinguish clearly between different AI-related evaluation scenarios, even though the literature shows that environmental impacts and performance characteristics can differ substantially across them [9, 33,34,36,39]. Xu et al. explicitly discuss training and inference as distinct contexts of computation and recommend scenario-sensitive reporting measures such as FLOPs, running time, model size, and carbon emissions for fair comparison [33]. A synthesized definition should therefore incorporate AI-related scenarios, such as training, inference, and refinement, as evaluation-relevant abstractions so that Green AI claims can be interpreted relative to comparable modes of system development and use. This is not meant to prescribe a specific lifecycle model but to distinguish functionally different evaluation contexts for more transparent and comparable assessment.

Performance preservation and functional comparability. Existing Green AI definitions do not frame efficiency as an isolated objective. Schwartz et al. define Green AI as research that yields novel results while taking computational cost into account, Xu et al. restate this orientation as obtaining novel results without increasing computational cost and ideally reducing it, and Patterson et al. characterize Green AI as focusing on computing efficiency as well as model quality [14, 33,34]. Taken together, these formulations imply that Green AI cannot be reduced to minimizing resource use alone. A synthesized definition should therefore preserve functional comparability by relating environmental improvement to maintained or improved task performance rather than to cost reduction in isolation.

This also affects how Green AI claims can be compared. Xu et al. explicitly discuss the need for fair comparison and recommend reporting measures such as FLOPs, running time, model size, and carbon emissions [33]. Schwartz et al. likewise argue that efficiency should be evaluated alongside accuracy and related measures [14]. Henderson et al. call for systematic reporting of energy and carbon metrics to enable more consistent comparisons and better reproducibility [40]. In a related way, Tornede et al. argue in the AutoML context that final performance values alone are insufficient and propose comparing approaches through performance profiles that relate achieved quality to the consumed computational budget [36]. A synthesized definition should therefore require that environmental gains be interpreted relative to a declared functional objective and comparable evaluation setting so that efficiency claims remain meaningful in relation to the level of performance achieved.

From compute efficiency to environmental impacts. Existing Green AI definitions initially frame the problem primarily in terms of computational cost or computing efficiency. Schwartz et al. define Green AI as research that yields novel results while taking computational cost into account, Xu et al. restate this as obtaining novel or comparable results with as few computations as possible, and Patterson et al. summarize Green AI as focusing on computing efficiency as well as model quality [14,33,34]. In these formulations, environmental relevance is clearly present, but it is often expressed through compute-related proxies such as computational cost, efficiency, runtime, FLOPs, or energy use rather than through a broader environmental-impact perspective.

More recent contributions broaden this perspective by framing Green AI more directly in terms of environmental impacts. Chen et al. argue that Green AI highlights the need to take environmental impacts into major consideration when developing AI, while Bolón-Canedo et al. explicitly refer to environmental cost and carbon footprint in model design, training, and deployment [10,35]. Verdecchia et al. broaden the scope by including both mitigation of AI’s own environmental impact and the use of AI to mitigate human environmental impact [9]. These contributions shift the focus from computational efficiency alone to environmental impacts more generally. A synthesized definition should therefore treat compute-related proxies as indicators, but define the minimization target in terms of environmental impacts.

From single-impact to multi-impact assessment. Li et al. demonstrate that AI models can have a substantial water footprint beyond carbon alone, while Gupta et al. show that embodied emissions from hardware manufacturing and infrastructure are becoming increasingly important relative to operational emissions [7,8]. A synthesized definition should therefore account for multiple environmental impact dimensions, because AI systems may affect energy use, greenhouse-gas emissions, water use, and embodied or material impacts in different ways across their lifecycle.

Relation between conceptual definition and operational assessment basis. Existing definitions and formulations of Green AI differ not only in scope but also in their degree of operational clarity. In particular, the literature shows that environmental claims are often made at a conceptual level, whereas the practical basis for assessment remains underspecified regarding system boundaries, indicators, measurement procedures, and reporting assumptions [17,22,40]. Measurement and evidence are therefore essential for operationalizing Green AI, even if the definition itself remains conceptual in form.

Trade-offs, uncertainty, and reference frame. Existing Green AI definitions and related formulations often remain underspecified regarding how environmental claims should be interpreted relative to competing objectives, methodological assumptions, and comparison settings. This is problematic because the literature repeatedly shows that environmental assessments of AI depend on explicit reference frames, including the choice of metric, infrastructure, region, timing, and comparison baseline [14,17,34,40].

A first recurring issue concerns trade-offs. Schwartz et al. argue that efficiency should become an evaluation criterion alongside accuracy and related measures rather than replacing them [14]. Henderson et al. similarly recommend considering energy-performance trade-offs before deploying energy-hungry models [40]. Patterson et al. show that substantial reductions in energy use and CO₂e can be achieved while maintaining quality but also make clear that meaningful comparison requires stating the relevant conditions under which such claims are made [34]. A synthesized definition should therefore not treat lower environmental impact as an isolated target but interpret it relative to declared performance objectives and comparable evaluation settings.

A second issue concerns uncertainty and methodological variability. Dodge et al. show that operational carbon intensity depends materially on data center region and even time of day, while Bannour et al. report that different carbon estimation tools can yield significantly different results for similar NLP experiments [17,41]. Patterson et al. likewise emphasize that carbon estimates depend on assumptions such as processor type, data center efficiency, and carbon intensity of the energy supply [34]. A synthesized definition should therefore require that Green AI claims be interpreted against an explicit reference frame, including the declared comparison object, functional perspective, and key accounting assumptions, while leaving the detailed reporting protocol to the measurement section.

Building on the recurring definitions and shortcomings identified above, the following synthesized definition of Green AI is formulated.

Synthesized definition of Green AI from the current body of knowledge.

Green AI refers to AI systems whose cumulative environmental impacts are reduced across declared system and lifecycle boundaries while maintaining comparable functional performance. Its assessment requires a system-level, lifecycle-aware, and multi-impact perspective under explicit reference conditions and transparent assumptions.

This synthesized definition captures the conceptually narrower common ground distilled from the reviewed definitions and shortcomings, while the following subsection makes explicit those additional interpretive elements that remain underspecified in this current body of knowledge.

4.1.2. Review-based analytical completion of the synthesized definition

The following aspects are not specified consistently in the reviewed Green AI definitions, but emerge as analytically relevant when completing unresolved definitional issues exposed by the literature.

AI-task fulfillment and realized use context. The reviewed literature shows that the environmental impacts of AI are not exhausted by model training alone but arise in concrete contexts of development, deployment, operation, monitoring, and use. Patterson et al. distinguish between research, development, and production settings and emphasize that operational impacts depend on how models are actually run. Ligozat et al. similarly argue that AI services must be assessed beyond model computation alone, including data acquisition, transfer, storage, processing, and use. Verdecchia et al. also stress that Green AI requires attention to multiple lifecycle stages and artifacts, such as data, models, pipelines, architectures, and hardware [9,22,34].

This suggests that Green AI cannot be grounded sufficiently in abstract model descriptions alone but must also consider the realization context in which AI capabilities are instantiated for concrete task fulfillment. In this sense, environmental impacts depend not only on what model is used but also on how, where, and how often it is executed. For example, operational profiles may differ depending on deployment architecture, inference frequency, and runtime management in concrete usage settings. On this basis, it is analytically useful to complement the synthesized definition with an explicit focus on AI-task fulfillment and realized use context. This provides a conceptual bridge between AI system realization and scenario-based evaluation because it links

system design to the observable environmental effects that arise in concrete modes of development and use.

Reference states, targets, or thresholds. The reviewed literature repeatedly shows that Green AI claims require an explicit reference frame but does not provide one universally accepted target or threshold. Henderson et al. call for systematic reporting of energy and carbon metrics to enable comparison and reproducibility. Dodge et al. show that carbon intensity depends materially on region, timing, and accounting assumptions, while Patterson et al. recommend publishing emissions explicitly to foster competition on more than model quality alone [17,34,40]. Accordingly, a review-based analytical completion should make explicit that Green AI claims are interpretable only relative to declared reference states, such as baselines, performance targets, or externally stated benchmarks. Fixed thresholds should therefore be avoided because they are context-dependent and subject to technological change. Green AI claims are thus more appropriately interpreted relative to declared reference states than to one universal threshold. Such reference states can also support comparisons with non-AI alternatives or alternative AI system realizations. However, the literature does not support a single universal threshold for classifying AI systems as green across all contexts.

Context-dependent interpretation of Green AI claims. The reviewed literature further suggests that Green AI claims are context-dependent. Kaack et al. distinguish between computing-related impacts, immediate application effects, and broader system-level effects, thereby showing that the overall climate relevance of machine learning depends on more than model-internal efficiency. Wu et al. likewise analyze AI from a holistic perspective spanning data, algorithms, and hardware, while considering both operational and manufacturing emissions. Verdecchia et al. emphasize that Green AI spans multiple lifecycle stages and artifacts, and Li et al. show that even the water footprint of AI models varies across spatial and temporal operating conditions [8,9,23,32]. A review-based analytical completion should therefore clarify that Green AI claims must be interpreted relative to declared technical, infrastructural, and regional conditions rather than as context-free properties of a model or method.

State of best AI system realization. The reviewed Green AI definition literature does not itself provide a direct and established concept of a “state of best AI system realization”. This expression is therefore not introduced as a literature-derived term but as an analytical shorthand for the comparatively best available realization of an AI system under declared evaluation conditions. This interpretation is consistent with the reviewed literature insofar as Green AI is repeatedly discussed in relation to efficiency improvements, best practices, transparent comparison, and the need to relate environmental performance to maintained quality [14,34,40]. In this restricted analytical sense, the term denotes a comparative state rather than an absolute property and serves to express that Green AI should be assessed relative to currently achievable system realizations under explicit conditions.

Constraint context of realization. Existing Green AI definitions do not usually integrate economic, societal, or governmental constraints as explicit definitional elements. However, adjacent sustainability-oriented literature shows that AI systems are embedded in broader socio-technical and governance contexts. Kaack et al. discuss policy levers and system-level effects for aligning machine learning with climate mitigation, Wu et al. examine sustainable AI from a holistic perspective spanning technical and infrastructural dimensions, and Van Wynsberghe explicitly frames Sustainable AI in relation to environmental, social, and economic tensions across the lifecycle of AI products [23,32,42]. In addition, broader Green AI initiative literature points to regulation, infrastructure, and deployment constraints as practically relevant conditions for realizing greener AI systems [37]. A review-based analytical completion may therefore acknowledge a constraint context of realization but should do so cautiously: not as a core literature-based definitional criterion of Green AI itself but as a contextual condition affecting how greener system realizations can be

achieved in practice. In this sense, reassessment may become necessary when new technical alternatives become available or when the declared evaluation boundaries change.

Building on the synthesized definition of Green AI, the following proposed review-based analytical completion definition goes beyond descriptive synthesis by integrating normative, operational, and theory-building elements that are not consistently specified in the current body of knowledge.

Proposed definition of Green AI.

Green AI refers to a state of best AI system realization considering all research approaches, practices, and techniques to achieve, within the declared system and its AI lifecycle boundaries, the minimum of the AI's cumulative negative environmental impacts and the AI's maximum of its cumulative positive environmental impacts in comparable AI-related scenarios for AI-task fulfillment throughout the corresponding lifecycle of an AI, is evaluated relative to explicit reference conditions based on measured or calibrated evidence regarding published targets or thresholds with transparent trade-off rules with documented uncertainty, and is consistent with economic, societal, and governmental constraints.

While the synthesized definition captures the conceptually narrower common ground, the proposed review-based analytical completion definition extends it by introducing additional analytical constructs, a normative orientation, and an explicit operational perspective for evaluation and comparison.

Taken together, the synthesized definition and its review-based analytical completion separate the narrower definitional core from additional interpretive elements that are relevant for operational reading but are not specified consistently in the literature so far.

4.1.3. Limitations and open challenges

The synthesized definition and its proposed review-based analytical completion definition are subject to several limitations. First, the Green AI literature remains heterogeneous and still evolving, with a strong concentration on energy-efficiency-oriented formulations and comparatively fewer studies addressing broader environmental footprints, emissions estimation, or operational assessment in a more systematic way [9]. As a result, any synthesized definition necessarily reflects a field in consolidation rather than a mature and fully stabilized conceptual consensus.

Second, the reviewed literature does not yet provide uniformly established system boundaries, indicators, or assessment conventions. Existing approaches differ in how they delimit the assessed system, which lifecycle stages they include, and which proxies or accounting assumptions they adopt [17,22,40]. This affects not only the comparability of Green AI claims but also the degree to which a conceptual definition can be operationalized consistently across studies, infrastructures, and application settings.

Third, important parts of the environmental footprint remain difficult to quantify in practice. Several contributions show that carbon estimates depend materially on regional and temporal electricity data, infrastructure assumptions, and methodological choices [17,34]. More broadly, adjacent assessment-oriented work stresses that relevant impacts may extend beyond operational energy use to data acquisition, transfer, storage, hardware production, and water consumption, while some of these components remain difficult to isolate, measure, or attribute reliably [8,22,36]. Consequently, even operationally informed Green AI definitions still face non-trivial measurement and attribution limits.

Fourth, the literature also points to limited tool maturity, methodological variability, and unstandardized reporting practices. Verdecchia et al. note that Green AI tool provisioning is still scarce, while Alzoubi and Mishra emphasize data-quality dependence, methodology variability, limited scope, and the need for more transparent and standardized assessment practices across tools and initiatives [9,37]. Henderson et al.

likewise observe that energy and carbon metrics are still not reported consistently across machine learning studies [40]. This means that the practical interpretability of Green AI remains partly constrained by the state of measurement and reporting infrastructures.

Fifth, practical implementation of Green AI measures may itself be constrained by regulatory, economic, or organizational conditions, even where the conceptual direction of improvement is clear [23,32,42]. Such constraints affect feasible optimization choices in practice but should not be treated as part of the core definition itself.

Finally, the proposed Green AI definition introduced above includes interpretive elements that are not specified consistently in the reviewed Green AI definitions themselves. In particular, notions such as reference states, AI-task fulfillment, context-dependent interpretation, or a constraint context of realization are introduced here as analytical completions of unresolved definitional issues rather than as uniformly established literature-based elements. They help specify how Green AI claims can be interpreted operationally and where further research is needed.

4.2. Transition to the Green AI lifecycle

This section develops a review-derived analytical lifecycle structure based on the synthesized Green AI literature. To address the identified gap between isolated term usage and a more holistic lifecycle perspective, the analysis moves from term-level evidence to phase-level synthesis (cf. Online Appendix, Section D). The resulting structure covers provisioning and siting, model development, training, deployment/inference, maintenance, and end-of-life contexts, together with the environmental impact dimensions discussed across the reviewed literature, including energy, CO₂e emissions, water use, and embodied/material footprints [40,41,43].

To characterize the state of the art, a 43-study subcorpus was analyzed using the six-step synthesis procedure summarized below and documented (cf. Online Appendix, Section D). First, lifecycle-related terminology was extracted using keyword-in-context analysis and synonym consolidation to obtain a harmonized vocabulary. Second, quantitative term analysis was conducted using length-normalized frequencies and a composite salience ranking. Third, coverage coding to construct a binary source-criterion matrix and corresponding heatmaps. Fourth, hierarchical clustering of term co-occurrence to identify relationships between lifecycle phases. Fifth, phase-level mapping and a coverage matrix are used to aggregate harmonized terms into review-derived subphases and phases. Sixth, statistical analysis of lifecycle coverage is used to quantify phase-level imbalances and overall lifecycle breadth [26–28,31].

The following analysis addresses RQ2: “Which phases and subphases structure the *Green AI lifecycle*, and how can they be analytically related to the stages of Life Cycle Assessment (LCA)?”

First, it reports which lifecycle phases are attested in the corpus. Second, it synthesizes them into a scheme of five phases and 33 subphases aligned with the dendrogram leaf order (cf. Online Appendix, Figure D.6). The shared leaf order provides a fixed column basis for the phase aggregation (cf. Online Appendix, Table D.3) and for quantifying the resulting imbalances. The identified subphases should therefore be interpreted as *analytical categories derived from the literature*, not as prescriptive process steps. The 33 subphases are used as review-derived coding categories to structure the literature corpus rather than as a universally validated lifecycle standard. Their level of granularity follows from the applied methodology, including terminology extraction, coverage coding, and hierarchical clustering, which required a sufficiently fine-grained analytical structure to capture differences in how lifecycle activities are addressed across studies. In particular, the use of a source-criterion matrix and phase-level coverage analysis relies on this granularity to make variations, imbalances, and underrepresented lifecycle aspects systematically visible. In step 5, a subphase is coded only when it is evidenced by operational content,

a quantitative metric with unit and declared scope/provenance, or an implemented procedure with reproducible configuration/results. At the same time, the subphases are not intended as an operational blueprint for implementation but as analytical structuring elements derived from the review. Accordingly, they can be aggregated into higher-level groupings or adapted in their level of detail depending on the application context without changing the underlying analytical structure. Third, it quantifies coverage and imbalances at phase-level granularity. This quantification uses criterion prevalence, per-phase mean coverage, and overall lifecycle breadth. Fourth, it consolidates the results into a review-derived analytical lifecycle structure.

Using the harmonized vocabulary, salience ranking, and the chronological heatmap, lifecycle phases are identified in the corpus. The term-level basis for this aggregation is provided (cf. Online Appendix, Table F.7), and the resulting phase-level coverage matrix is reported (cf. Online Appendix, Table D.3). These signals are synthesized into a scheme of five phases with 33 subphases used in the synthesis that are aligned to the dendrogram leaf order (cf. Online Appendix, Figure D.6) and the coding rules used to support systematic coverage comparison. Thereafter, phase-level coverage and imbalance (criterion prevalence, mean coverage per phase, overall breadth) were quantified, identifying over- and underrepresented areas based on the matrix. Finally, Section 4.2.1 presents the resulting review-derived lifecycle structure in analytically condensed form.

This staged exposition connects term-level evidence to phase-level synthesis and quantified coverage, yielding a review-derived analytical lifecycle structure with explicit system and lifecycle boundaries, analytically derived phase categories, and interpretable reporting dimensions that support lifecycle-oriented interpretation and comparison. Together, steps 5 and 6 connect raw terminology extraction with structured phase-level coverage analysis.

4.2.1. A synthesized Green AI lifecycle structure

Based on the structured analysis of the reviewed literature, the following review-derived lifecycle structure is presented as a *synthetic abstraction*. It does not represent a directly observed standard process but consolidates recurring patterns, activities, and evaluation situations identified across studies into an analytical structure.

The aggregated results reveal strong imbalances. Deployment-related subphases (*efficient model training and testing*, *energy-aware inference*) exhibit the highest mean coverage rates, reaching up to 39.5%. Hardware-oriented subphases — *responsible material sourcing* and *green logistics* — achieved 36.7% coverage, exceeding *Green AI development* (24.0%) and *circular AI maintenance* (22.1%), while *green end-of-life* and *AI circularity* remained lowest at 17.1%. These metrics are consistent with the term-frequency analysis from step 2 and suggest that lexically underrepresented topics also appear less structurally developed in the reviewed lifecycle coverage. Low-salience subphases, such as *circular recycling* and *green compliance*, indicate underrepresented areas for future Green AI research and policy-oriented discussion. The resulting lifecycle synthesis addresses this fragmentation by integrating upstream, core compute, and downstream activities into a structured view of the *Green AI lifecycle*. In this way, the five high-level Green AI phases and the set of subphases identified in the literature are related within a coherent analytical structure (cf. Online Appendix, Table D.3).

In this sense, the lifecycle synthesis contributes not only a structured abstraction of dispersed lifecycle terms but also a quantified view of where the literature systematically over- and underrepresents Green AI-relevant lifecycle activities.

The identified subphases should therefore be interpreted as *analytical categories derived from the literature*, not as prescriptive process steps. Likewise, the introduced constructs are used as review-derived structuring elements rather than as independent frameworks for direct implementation. Their aggregation follows the dendrogram-based structuring and the phase-level coverage matrix (cf. Online Appendix, Figure D.6).

The lifecycle synthesis is illustrated at three complementary levels. First, a Level-0 view relates the synthesized phases to an LCA-oriented perspective. Second, Fig. 1 presents an illustrative notation-based view of the five synthesized phases and their analytical relation. Third, more detailed views are provided in the Online Appendix (Section H.2).

Taken together, these visualizations present the review-derived lifecycle structure at different levels of abstraction. The notation-based figure should be read as one possible visual arrangement of the synthesized categories, not as a prescriptive process model, a validated lifecycle logic, or evidence of the superiority of a particular notation.

Analytical function of the review-derived lifecycle structure.

An analytical lifecycle perspective is used here to describe Green AI-relevant activities, evaluation situations, and possible improvement actions across the lifecycle. In the synthesized structure, measurement- and evaluation-related terms recur across several lifecycle phases and link provisioning and siting with deployment and operation. Upstream and downstream subphases appear as part of the same analytical structure rather than as peripheral elements (cf. Online Appendix, Section D.3). The review-derived lifecycle structure is not intended as a governance framework but as an analytical structuring of lifecycle-related activities, evaluation situations, and trade-offs across performance, carbon, water, and embodied environmental impacts.

Illustrative follow-up situations and notation-based figure interpretation. In the notation-based view, *adapt* and *adjust* denote two illustrative types of follow-up situations. *Adapt* refers to changes in the operational context, whereas *adjust* refers to changes in the AI artifact or its immediate configuration. Fig. 1 shows the review-derived Green AI lifecycle categories at the selected level of aggregation. It distinguishes neutral control flow, progression flow after gate evaluation, and rework flow after gate evaluation, and includes the follow-up situations *adapt* and *adjust*. The figure is a reading aid for the synthesized categories. It should not be read as a prescriptive process model, a validated lifecycle logic, or evidence that one notation is superior.

Relating the review-derived Green AI lifecycle phases to LCA supports a more systematic interpretation of environmental impacts and improves compatibility with established reporting schemes [22,40]. The linkage relates AI-related activities, from workload scheduling to hardware recovery, to established LCA concepts. It thereby supports the interpretation of carbon-, water-, and material-related impacts across lifecycle phases. Table 1 summarizes this relation across the aggregated lifecycle phases. Rather than implying a direct one-to-one correspondence, the mapping shows how the five review-derived lifecycle phases are analytically related to LCA stages at the Level-0 aggregation. It serves as an analytical reading aid rather than implying equivalence between lifecycle activities and LCA phases.

A more detailed subphase-level analytical reading aid is provided in the Online Appendix (Section G), where the review-derived lifecycle categories are analytically related to LCA stages. The mapping does not assign phases to single LCA stages but rather illustrates typical points of interaction across stages.

4.3. Green AI hardware

To address RQ3 (“Which hardware components and system architectures are associated with improved performance-per-watt and potentially reduced embodied environmental impacts across the lifecycle?”), hardware used in Green AI research was synthesized. Detailed material is provided in the Online Appendix (Appendix I). Building on this synthesis, a criteria-based selection perspective is introduced that remains applicable across hardware generations. Section 4.3.1 then makes spatial and temporal carbon-intensity factors explicit and outlines analytical reporting dimensions. Finally, Section 4.3.2 provides an illustrative class-based overview that reflects common deployment classes.

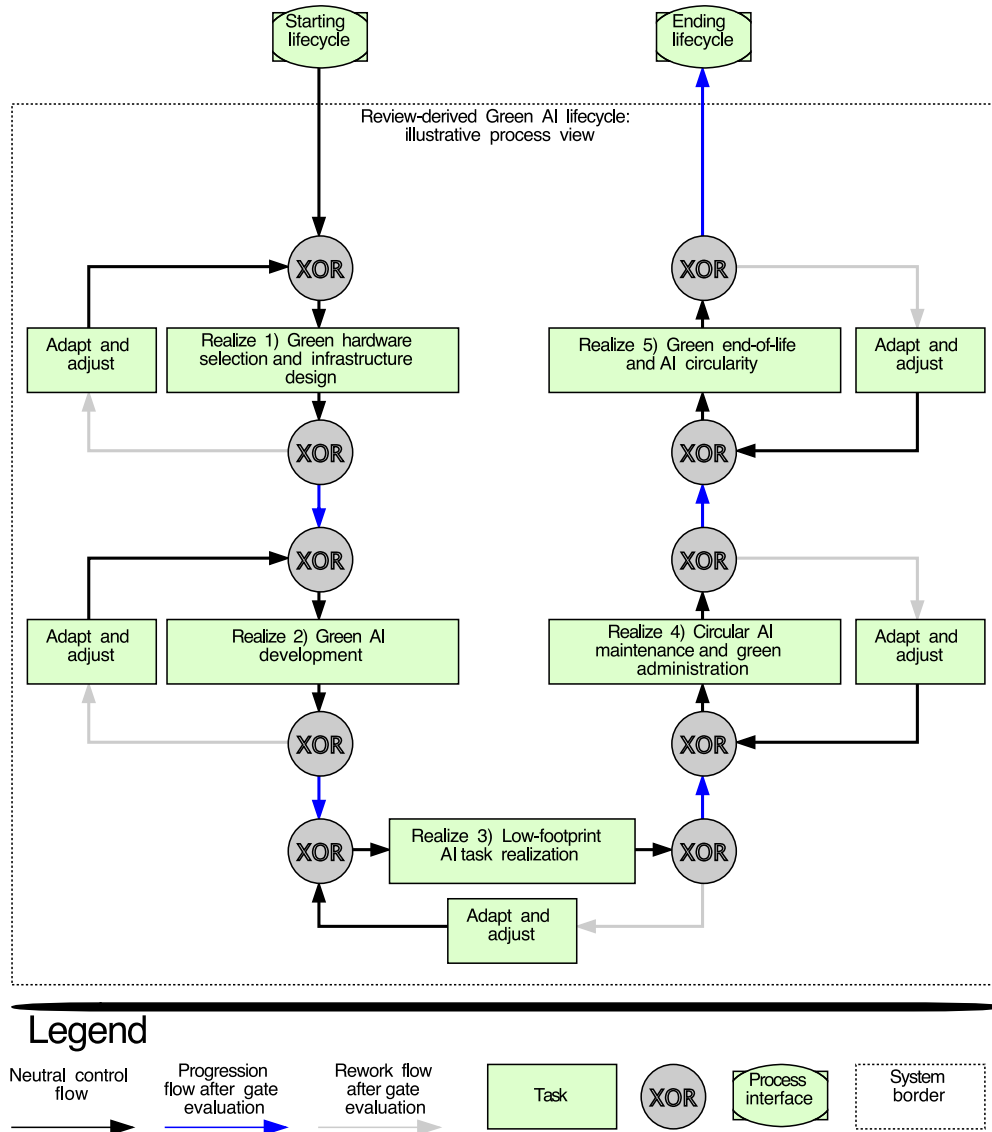


Fig. 1. Illustrative notation-based transparency view of the review-derived Green AI lifecycle categories. The figure visualizes the five synthesized phases together with neutral control flow, progression flow after gate evaluation, rework flow after gate evaluation, and the illustrative follow-up situations *adapt* and *adjust*. As an illustrative example, the lifecycle was visualized using notation elements inspired by the Neuronal Modeling and Description Language (NMDL) [24]; alternative workflow-oriented representations would be equally feasible.

Table 1

Phase-level mapping of the review-derived Green AI lifecycle to LCA stages. The mapping provides an analytical orientation and does not imply that lifecycle phases are identical to LCA stages.

Green AI lifecycle phase (Level-1)	Illustrative LCA linkage/typical assessment relevance (Level-0)
(1) Green hardware selection and infrastructure design	Related to goal and scope definition and inventory analysis, especially through system boundaries, hardware choices, and embodied impacts. These aspects also influence later impact assessment and interpretation.
(2) Green AI development	Associated with goal and scope definition and inventory analysis. Development choices shape how environmental indicators are derived and interpreted, particularly when trade-offs between performance and impact are considered.
(3) Low-footprint AI task realization	Closely connected to inventory analysis and impact assessment. Deployment and inference produce operational resource flows that are quantified as environmental indicators and assessed under performance constraints.
(4) Circular AI maintenance and green administration	Primarily linked to interpretation, supported by monitoring, benchmarking, and compliance activities. These activities can inform subsequent inventory and impact assessment over time.
(5) Green end-of-life and AI circularity	Related to inventory analysis, impact assessment, and interpretation. Decisions on reuse, recovery, recycling, and disposal also influence system boundaries defined in earlier stages.

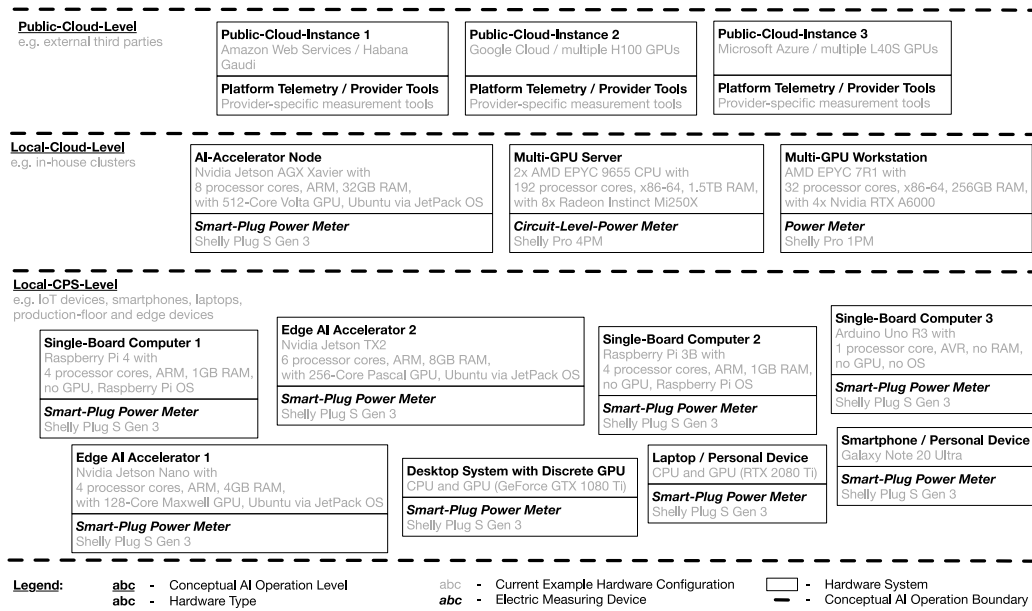


Fig. 2. Illustrative class-based overview of representative hardware systems and measurement contexts discussed in the Green AI literature across local CPS, local cloud, and public cloud levels.

4.3.1. Contextual determinants of Green AI hardware

Operational emissions are materially affected by *where* and *when* workloads run [40,43]. (i) Regional electricity mixes can shift carbon intensity by an order of magnitude; location and emission sources should be reported [40]. (ii) Temporal variation is relevant: carbon-aware scheduling aligns execution with renewable availability while respecting SLAs [43]. (iii) Facility overhead (PUE, cooling) contributes substantially and should be captured via whole-system measurements where feasible [40]. (iv) Embodied environmental impacts from production and replacement should complement operational indicators; longer duty cycles and modular, repairable systems can reduce amortized impacts, depending on utilization, efficiency gains, and replacement assumptions [8,22]. (v) Data locality and I/O externalities can shift the greener option between on-premises and cloud deployments [41,44].

Analytical reporting dimensions. Across the reviewed studies, hardware-related environmental results depend strongly on contextual variables such as location, Power Usage Effectiveness (PUE), carbon-intensity source and temporal granularity, scheduling window, measurement method, workload profile, and embodied-carbon assumptions [40,43]. These variables explain why hardware comparisons remain difficult across studies and why reported results are only interpretable within clearly specified infrastructural contexts. Adjacent work on IIoT platform selection similarly treats platform choice as a socio-technical decision problem spanning infrastructure, application functions, runtimes, and sustainability-related criteria such as energy consumption and CO₂ emissions [45].

Class-based synthesis. The reviewed literature does not support device-level prescriptions but indicates recurring deployment classes across the edge-cloud continuum [38,41,46–50]. This mapping includes: (i) low-power single-board computers or edge accelerators for latency-sensitive inference at the local Cyber-Physical System (CPS) level; (ii) workstations for mid-scale training or inference where moderate local resource constraints dominate; (iii) on-premises servers where control over PUE, data locality, and system configuration is relevant; and (iv) public cloud accelerators for bursty or compute-intensive workloads, where regional and temporal scheduling can affect environmental performance [38,40,41,43,44,46–50]. The configurations in Online Appendix (Table I.9 and Table I.10) illustrate such recurring classes and representative hardware heterogeneity discussed in the literature. In

the figure, the upper labels indicate the general class- and deployment-level logic, whereas the lower gray entries show current example configurations and their corresponding measurement-device mappings. These concrete mappings are included as an illustrative snapshot of currently relevant Green AI hardware and measurement setups and may require updating as hardware generations and measurement options evolve [9,10,35,38,41].

4.3.2. Illustrative hardware systems

Fig. 2 provides an illustrative class-based view of hardware systems discussed in the Green AI literature across conceptual AI operation levels adapted from Grum et al. [51], namely the local CPS level, local cloud level, and public cloud level. The figure combines a class-based comparative perspective with current example configurations and corresponding measurement-device mappings to make heterogeneous hardware and measurement contexts more interpretable.

In the literature, *edge-centric* or *small-scale AI applications* are often associated with energy-frugal hardware platforms. Low-power edge accelerators and single-board computing platforms combine low idle power draw with integrated inference capabilities and therefore recur primarily in studies of latency-sensitive and locally executed workloads [20,39,47,52]. The reviewed literature places specialized edge accelerators in local CPS settings, where lower data movement can reduce communication-related overheads. Laptops, smartphones, and desktops used in consumer and Small and Medium Enterprise (SME) contexts are treated in the same class-based view because similar deployment constraints apply [22,36,40,41,43]. Across these settings, the studies consistently describe tensions between energy efficiency, compute capacity, and infrastructure coupling. In such local CPS settings, device-level or plug-based metering is often comparatively feasible, which makes the relation between hardware type and measurement approach especially visible.

The reviewed literature places locally operated servers with CPUs and mid-range accelerators at the local-cloud level when addressing medium-scale requirements. In these settings, the relevant analytical trade-offs involve compute capacity, energy demand, operational cost, and maintainability [10,23,35]. Further, powerful workstations can also be positioned on this level, as they can combine multiple non-server GPUs with dedicated cooling requirements and comparatively moderate requirements on the available electric power supply

infrastructure. The local cloud level is treated as a review-derived deployment class that reflects typical trade-offs between compute capacity, energy demand, and infrastructure dependency in medium-scale Green AI settings. At this level, measurement typically shifts from simple device-level metering toward node-, circuit-, or server-level measurement contexts.

The reviewed literature addresses hyperscale cloud accelerators mainly in compute-intensive training and large-scale inference settings [23,34,50]. From a Green AI perspective, their significance is determined less by individual device properties than by infrastructure-level conditions, such as utilization, cooling, interconnect overhead, and power availability [53,54]. In this sense, the public-cloud level constitutes a distinct deployment context with characteristic trade-offs between throughput, infrastructure dependency, and environmental overhead. These settings also rely on specialized orchestration software, including distributed training frameworks and cluster schedulers. Because direct hardware-level instrumentation is often unavailable in public-cloud environments, measurement in these settings commonly relies on platform-specific telemetry and provider-side tools rather than on device-level metering.

From a Green AI perspective, the critical factor is not the specific device model but the efficiency characteristics of the deployment architecture, as reflected in the abstracted hardware classes presented above [22,23,41,55,56]. In this sense, the synthesis contributes a class-based view of heterogeneous hardware contexts that highlights trade-offs between energy efficiency, compute capacity, infrastructure coupling, and measurement context.

4.4. Green AI measurement tools

Assessing whether an AI system meets the Green AI definition requires reliable and traceable measurement approaches. Building on the Green AI definition (Section 4.1), the synthesized Green AI lifecycle (Section 4.2), and the class-based hardware perspective (Section 4.3), this section examines how environmental impacts can be measured across AI systems. Robust measurement is necessary to assess Green AI claims consistently and to make results comparable across workloads, infrastructures, and lifecycle stages.

To answer RQ4 (“How can environmental impacts (e.g., energy, CO₂e emissions, water use, and embodied/material footprints) be measured and reported consistently across the lifecycle, including calibration of indirect estimators with metered or telemetry-based reference data?”), this section synthesizes the measurement approaches identified in the reviewed literature and contrasts indirect estimator libraries, such as *CodeCarbon* or *Carbontracker*, with direct hardware telemetry. Relevant telemetry sources include Performance Monitoring Counters (PMCs), rack-level Power Distribution Units (PDUs), and facility-level monitoring. The reviewed evidence shows that estimator-based approaches provide broad coverage and low integration overhead, whereas metered or telemetry-based approaches can provide higher-fidelity reference data within specified measurement boundaries but remain less integrated into AI-specific tooling. This tension motivates a hybrid measurement logic in which software-based estimates are calibrated against hardware-level observations and reported with explicit assumptions and uncertainty. It also shows that measurement validity depends on workload profile, system boundary, and reporting purpose rather than on tool choice alone. Supplementary analysis of the reviewed tool landscape further supports this pattern. Of the 16 identified tools, Python packages dominate (8/16), indicating the practical prevalence of easily applicable estimator-based approaches. Corporate dashboards and web-based tools are less common (3 each), while the remaining tools fall into smaller categories. However, only a few tools extend to hardware-level telemetry or embodied impacts, despite their relevance for lifecycle-oriented Green AI assessment [6,36,53]. The current tool landscape therefore remains fragmented between broad but partial estimator-based coverage and more accurate yet less accessible direct measurement approaches. Further details are provided in the Online Appendix (Appendix J.1; Table J.11).

4.4.1. A review-derived selection logic for Green AI measurement

Building on the critical gaps identified above, the reviewed literature suggests a selection logic for Green AI measurement rather than a single preferred tool. The purpose of this synthesis is to structure how heterogeneous methods can be combined according to workload characteristics, system boundaries, and reporting objectives, while addressing the still limited cross-layer integration of current measurement approaches.

Dual-layered measurement logic. Fig. 3 summarizes a review-derived hybrid measurement logic that combines (i) *software-level estimators* for broad coverage and (ii) *hardware-level monitors* for calibration and validation. The software layer supports accessibility, automation, and workflow integration (e.g., *CodeCarbon*, *Carbontracker*), whereas the hardware layer provides metered or telemetry-based reference data for calibration and validation (e.g., smart plugs, PMCs, and rack-level PDUs).

Methodological differentiation. Recent comparative review work shows that measurement reliability in Green AI research depends strongly on methodological choices, including boundary-setting, calibration, and reporting practice [9]. This dependency underscores the importance of explicit system boundaries, transparent assumptions, and cross-layer verification for credible Green AI assessments. The reviewed approaches can be systematized along two dimensions: *indirect versus direct measurement* and *system- versus component-level granularity*. Indirect approaches include software-based estimators and provider-reported cloud dashboards, whereas direct approaches include external meters, rack-level PDUs, and integrated telemetry or sensor-based measurements. Because each approach has characteristic limitations, the reviewed literature points toward hybrid measurement setups that combine estimator portability with metered or telemetry-based reference data. A detailed comparison of measurement methods is provided in the Online Appendix, Table J.12.

This layered view makes the trade-off between scope and accuracy explicit and clarifies why indirect methods alone are often insufficient for robust Green AI reporting across system layers.

Selection principles. Synthesizing the reviewed literature, four recurring steps can be distinguished for selecting and combining measurement approaches. In *step (1) “define workload and scope”*, the workload is characterized (e.g., training versus inference, edge versus cloud, real-time versus batch), including its compute, memory, and I/O profile, service-level constraints, and duty cycles, and the required resolution is specified (e.g., system-level totals versus component-level attribution).

In *step (2) “select a baseline estimator”*, a software package or provider dashboard is used to establish broad measurement coverage and reproducibility. Assumptions should be documented explicitly, including emission-factor datasets, grid carbon data, location, temporal granularity, infrastructure characteristics such as Power Usage Effectiveness (PUE) where available, and hardware-related proxy values such as Thermal Design Power (TDP).

In *step (3) “add calibration points”*, direct hardware measurements are added to validate and refine baseline estimates. Depending on the setup, these measurements can include smart plugs or rack-level PDUs for whole-system load and PMCs for component-level signals. This is particularly important for capturing non-accelerator loads and overheads that remain invisible in GPU-centric estimates. Where available, benchmark results such as *Perun* or the *ML Energy Benchmark* can serve as additional plausibility references [36].

In *step (4) “extend beyond carbon”*, further indicators are incorporated where feasible, including water-related indicators, lifecycle amortization, and broader material or embodied-impact extensions. Although current water-use estimation models remain highly uncertain [8], reporting such dimensions explicitly helps expose current measurement gaps and improves cross-study transparency.

Illustrative workflow. For illustration, consider a training workload executed on an on-premises GPU cluster. A software-based estimator such as *CodeCarbon* can be used to record runtime, hardware

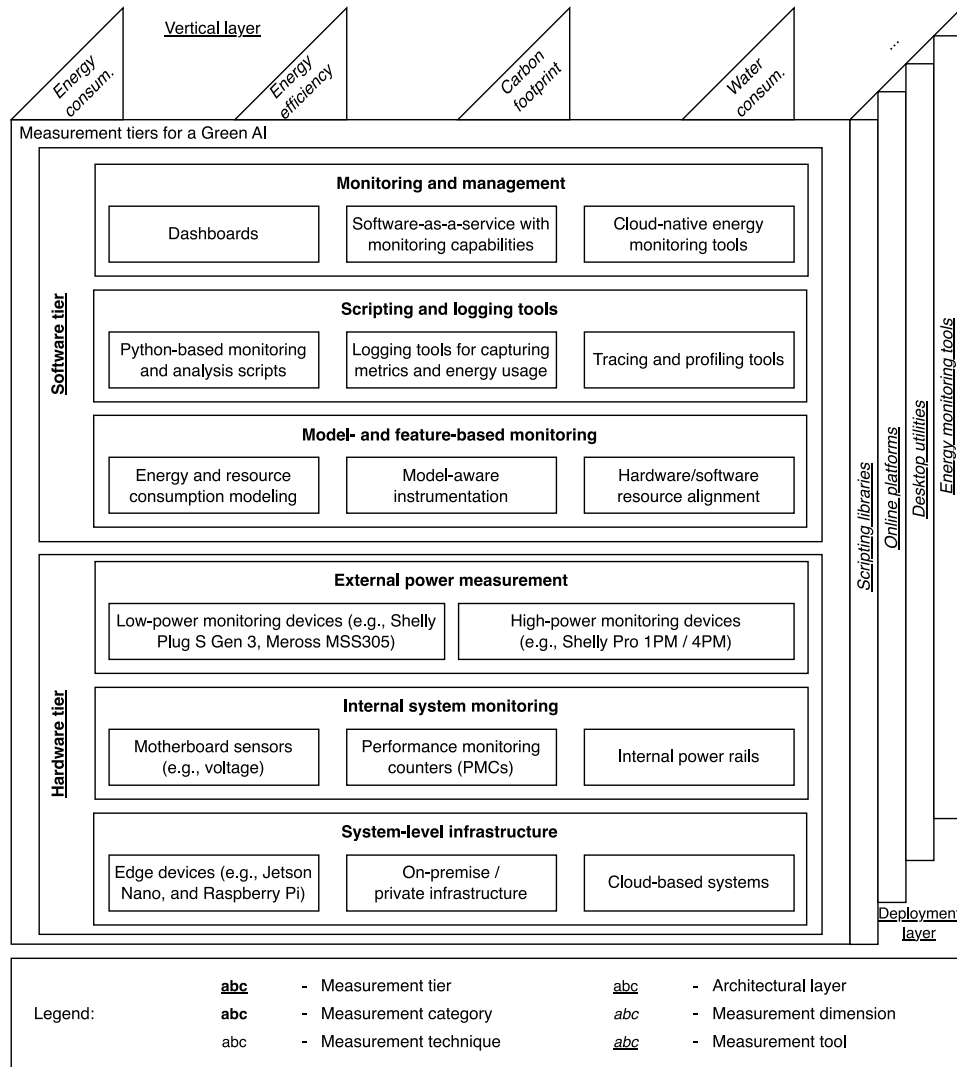


Fig. 3. Green AI measurement logic.

metadata, and estimated emissions, while rack-level smart plugs or PDUs capture measured power draw at the system level. Integrated PMCs and vendor-specific accelerator telemetry can complement this view with component-level signals for CPUs, GPUs, and memory where available. Reported results should document discrepancies between estimator outputs and metered values and disclose how calibration factors or emission-factor assumptions were applied in subsequent runs. This example illustrates how estimator coverage and hardware-based validation can be combined within a single measurement setup.

Synthesis outcome. The main value of this synthesis lies less in introducing new tools than in clarifying the structural tensions of the current measurement landscape. First, it shows that Green AI measurement is still skewed toward software-level convenience: indirect estimators scale easily across machine-learning workflows but remain sensitive to assumptions such as TDP values, utilization proxies, and emission-factor datasets. Second, it connects AI-focused estimator tools with hardware- and infrastructure-level monitoring practices established in Green IT and HPC, including performance monitoring counters, rack-level power metering, and facility-level telemetry, thereby making cross-layer calibration a central integration problem rather than a secondary refinement. Third, it highlights that reporting remains too narrow when it stops at energy and carbon, since water-related and lifecycle-oriented indicators are still weakly operationalized in current toolchains. Taken together, these points support a more

transparent and comparable Green AI measurement logic based on explicit assumptions, calibrated measurements, and broader indicator coverage.

4.4.2. Limitations of the synthesized Green AI measurement logic

Although this synthesized hybrid logic helps structure the fragmented Green AI measurement landscape, it remains constrained by dataset availability, emission-factor heterogeneity, and limited support for water- and lifecycle-related indicators. The main limitations are summarized below.

Reliance on indirect methods. Software-based estimators such as *CodeCarbon* or *Experiment Impact Tracker* rely on generalized emission factors and TDP-based assumptions [40,57]. While these tools are useful for rapid deployment, their validity decreases under heterogeneous workloads or non-standard infrastructures. Calibration against hardware measurements is therefore often necessary, yet it is still rarely performed systematically.

External meters versus integrated sensors. External devices such as smart plugs or rack-level PDUs provide simple deployment and architecture-independent whole-system coverage and are therefore particularly useful in heterogeneous or distributed environments, but they offer limited component attribution. By contrast, integrated PMCs and on-board sensors enable finer-grained profiling [36], yet their availability, interfaces, and calibration quality differ across hardware platforms.

Neither approach alone provides both complete system coverage and robust component-level attribution.

Incomplete metric coverage. Most tools focus on energy and carbon, while cooling-related water use, Water Usage Effectiveness (WUE) disclosure, and broader lifecycle or material dimensions remain weakly supported. Recent attempts to estimate water consumption rely on indirect cooling models [8], but these approaches still lack validation across diverse data center contexts. This omission constrains the comprehensiveness of current Green AI measurement and reporting practices.

Fragmentation and comparability. Outputs often remain difficult to compare because studies use different emission factors, regional datasets, and reporting formats [9]. This heterogeneity undermines cross-study benchmarking and reproducibility. More standardized reporting formats and clearer disclosure of assumptions would improve comparability. Related evaluation research on digital platforms indicates that explicit, multidimensional indicator sets and maturity-sensitive comparison logic can improve assessment comparability, suggesting a useful adjacent perspective for Green AI reporting and governance [58].

Lifecycle blind spots. Few tools capture embodied environmental impacts from semiconductor fabrication, assembly, and logistics or account for reuse and recycling, even though LCA studies indicate that such effects can materially affect overall footprint results [6,13]. Without harmonized methods for amortizing embodied carbon, reporting remains biased toward short-term operational footprints.

In summary, the current measurement landscape remains fragmented, carbon-centric, and assumption-sensitive. Addressing these limitations requires advances in three areas: relevant improvements include calibration procedures that align software estimators with hardware-based measurements, indicator models that better capture water use and lifecycle-related dimensions, and interoperable reporting formats for cross-system comparability. These improvements are especially important when studies differ in system boundaries (e.g., device-only versus whole-system) or in how carbon intensity is represented (e.g., average versus time-varying regional data). Together, they would support more consistent and transparent Green AI evaluation.

Software-based packages (e.g., *CodeCarbon*, *Experiment Impact Tracker*) represent indirect, model-based estimation. They integrate easily into machine learning pipelines but rely on generalized emission factors and hardware assumptions, which can reduce accuracy under heterogeneous workloads [40,57]. Cloud-provider dashboards (e.g., Google or Microsoft) also rely on indirect reporting, although they are enriched by provider-side infrastructure data such as cooling, Power Usage Effectiveness (PUE), and localized grid-mix information; however, they remain constrained to proprietary ecosystems [59,60]. Online tools serve a different purpose. They can raise awareness and communicate approximate impacts, but they are usually too coarse-grained for system-level optimization or research-grade reporting [61, 62].

Direct measurement provides reference data only for the boundary that is actually measured. External meters can capture device- or rack-level power draw, but facility-level cooling overhead is included only when the measurement boundary covers the corresponding infrastructure [22]. Integrated sensors such as PMCs or motherboard energy probes enable finer-grained attribution to CPUs, GPUs, and memory, although their availability, interfaces, and calibration differ across hardware generations [36]. Vendor-specific APIs and heterogeneous sampling behavior further limit interoperability, so no widely adopted approach currently provides consistent cross-layer measurement across heterogeneous HPC, edge, and cloud environments.

Hybrid approaches. *IrEne* illustrates how model-specific energy prediction can complement measurement-based approaches, although shared calibration protocols remain limited [63]. At the same time, recent reviews stress that the absence of shared calibration protocols

and emission-factor disclosure still limits comparability across studies [9]. Progress therefore depends not only on additional tool support but also on open benchmarks and reporting protocols for assumptions such as location, temporal granularity, PUE, WUE, embodied-carbon amortization, and calibration practice. This would align AI-specific measurement more closely with established measurement and reporting practices from Green IT and HPC research.

This methodological differentiation shows that portability favors lightweight estimators, precision favors hardware metering, and transparency depends on explicit disclosure of assumptions, calibration, and system boundaries. Tool choice is therefore inseparable from granularity requirements, workload type, and deployment context. A more detailed conceptual mapping of Green AI measurement approaches along the two axes of measurement type and granularity is provided in the Online Appendix, Table J.13.

This conceptual distinction has practical implications. *Indirect system-level estimators* (e.g., *CodeCarbon*, *Experiment Impact Tracker*) can deviate markedly from metered values under heterogeneous workloads [40,53]. These deviations arise because estimator-based approaches typically abstract from parts of whole-system overhead, such as cooling, conversion losses, or data-movement effects, unless such factors are explicitly modeled or calibrated. Without calibration against measured data, reported footprints therefore remain sensitive to workload composition and infrastructure context.

Synthesis. Taken together, the reviewed literature indicates that portability, precision, and transparency are difficult to maximize simultaneously. Software-only approaches favor accessibility but remain assumption-sensitive, hardware-only approaches improve fidelity but limit portability and comparability, and hybrid approaches remain under-standardized. Advancing Green AI measurement therefore depends less on additional standalone tools than on better integrated, cross-layer measurement pipelines, explicit calibration routines, and clearer reporting of system boundaries, emission factors, and uncertainty. This is especially important for distributed and heterogeneous AI infrastructures, where system-wide measurement is essential for meaningful comparison.

5. Operationalization challenges and future research directions for Green AI

This section synthesizes challenges and research gaps identified in the systematic review and uses the lifecycle-oriented structure developed in the previous section as an analytical reference point. In line with RQ5 (“Which open *challenges and research priorities*, such as benchmarks, reporting schemas, uncertainty handling, circularity, and governance-related conditions, are needed to advance lifecycle integration of Green AI?”), this section identifies methodological blind spots that emerged from the reviewed literature. Examples include calibration protocols, embodied-impact accounting, and multi-metric integration. The section then derives future research directions for lifecycle-oriented Green AI measurement and reporting.

5.1. Challenges and research gaps in Green AI

Building on the Green AI definition (Section 4.1), adaptive task distribution across heterogeneous infrastructures emerges as one possible operationalization pathway, beyond efficiency improvements at the model level. Edge and on-device deployments can reduce data-transfer-related impacts when inference is performed locally, but their environmental benefits are constrained by limited accelerator memory, device heterogeneity, and inconsistent power-monitoring capabilities [9,50]. This suggests a research need for multi-criteria workload orchestration that considers environmental cost, latency, and resource utilization across distributed AI systems [51].

Flexible resource allocation. From a Green AI lifecycle perspective, operational variability is relevant because demand changes affect

energy use and emissions-related behavior. The reviewed literature provides only limited evidence of adaptive resource-management approaches integrated into operational AI workflows. Further work should therefore examine scheduling and resource-management algorithms that modulate AI workload execution, for example, by migrating tasks to platforms with higher performance per watt or by activating scaling mechanisms according to task demand. These strategies need to distinguish between tasks with different urgency levels and resource intensities [43,54].

Real-time analytics may require immediate processing, whereas deferrable tasks can sometimes be shifted to lower-carbon time windows or more efficient infrastructures. Such adaptive approaches must balance environmental objectives with performance, availability, and operational constraints [17,64]. In practice, Green AI assessment should account for performance, latency, availability, and deployment constraints. Resource-efficiency research also includes techniques such as reducing redundant gradient computations, minimizing computational overhead, and using mixed-precision arithmetic, although their energy and accuracy effects remain workload- and hardware-dependent.

Heterogeneous computing platforms. Heterogeneous platforms — from CPUs and GPUs to TPUs and low-power edge AI accelerators — can provide energy-reduction levers when workloads are matched to architectural strengths. These architectures differ in their suitability for control-heavy preprocessing, dense linear algebra, and high-throughput tensor operations. Edge devices can reduce data-movement overhead when inference is performed locally. A remaining research gap concerns runtime scheduling models that exploit these complementarities across architectures, minimizing energy per task and, where hardware utilization and lifetime assumptions are reported, reducing amortized embodied impacts per workload [47,65].

Carbon- and multi-impact-aware workload allocation. The reviewed literature indicates that real-time energy monitoring via software and hardware counters has been demonstrated, but validation against whole-system or rack-level meters is rarely reported, especially for heterogeneous workloads. Building on the methodological gaps identified for Green AI measurement (Section 4.4), a relevant future research direction is the development of scalable allocation models that consider both efficiency and the carbon intensity of the energy source. Examples include shifting eligible computations to lower-carbon data-center regions or, where local execution is more efficient, offloading selected tasks to performance-per-watt efficient edge devices [53]. A comparatively underexplored strategy in AI-specific workflows is carbon-aware scheduling, understood as deferring or relocating eligible tasks according to temporal and geographic variations in grid emissions [40,53,54]. Related mechanisms have been studied more extensively in Green IT and HPC research, where job schedulers consider energy availability and cost signals [65,66]. Their systematic integration into AI workflows could help connect “Green IT” scheduling practices with “Green AI”-specific measurement and optimization efforts. Beyond electricity-related carbon emissions, future work should examine how workload allocation can account for water use, embodied/material impacts, and technical trade-offs such as latency, availability, and data locality [8,16]. As a possible extension, allocation models could incorporate region-specific water-stress indicators and amortized embodied impacts, provided that assumptions about cooling water intensity, hardware utilization, and hardware lifetime are explicitly reported [8].

Methodological requirements for carbon-aware scheduling. Carbon-aware scheduling would typically require short-term carbon-intensity forecasts for eligible regions, explicit deferral windows linked to SLAs, and migration policies that limit additional data movement. For example, non-urgent training workloads may be shifted to lower-carbon periods, whereas latency-sensitive inference may be constrained to regions that combine low-carbon electricity with acceptable response times. Studies should therefore report permissible time shifts and the regional scope considered [40]. In practice, relocation policies need to

account for data gravity and compliance constraints, including egress costs and data residency, and should disclose non-migratable workloads or privacy-driven pinning decisions. Deferral windows aligned with day-ahead carbon-intensity forecasts may create emission-reduction potential for non-urgent training workloads, particularly where regional carbon intensity varies over time, while latency-sensitive inference remains more constrained.

Reporting requirements for carbon-aware studies. To improve comparability, carbon-aware studies would need to disclose several methodological elements. Relevant reporting elements include: (i) carbon-intensity regions, data sources, and temporal resolution; (ii) deferral windows and SLA constraints; (iii) migration policies and data-locality assumptions; (iv) telemetry sources and calibration error; and, where applicable, (v) objective weights for carbon, water, and embodied impacts. More consistent disclosure would improve cross-study comparability and could support the transfer of Green IT practices to AI-specific settings [40]. Where studies combine multiple environmental objectives, they should disclose whether trade-offs were handled through a weighted objective, Pareto analysis, or another decision logic.

Bridging to Green IT and HPC scheduling. Techniques from Green IT and HPC, including power capping, DVFS-based throttling, energy-aware job placement, and queue-level energy budgets, provide reference points for AI pipeline research. In AI settings, such mechanisms could inform research on cluster-level placement, power-capped training, checkpointing strategies, and data-locality-aware accelerator scheduling [65,66]. Preemption and periodic checkpointing can enable deferral within carbon windows, but their net environmental benefit depends on rollback overhead, checkpointing costs, workload duration, and SLA constraints.

Telemetry-informed calibration and scheduling. Where available, direct telemetry from PMCs and rack PDUs can be used to calibrate software estimators during deployment and to inform placement and deferral decisions. Reporting pre- and post-calibration error is necessary to assess whether calibration improves estimator reliability. For example, studies could report deviations between software-estimator outputs, such as *CodeCarbon* estimates, and rack-level power-meter readings. Such reporting should also indicate whether calibration changed workload allocation or siting decisions. Such reporting would bring AI-specific measurement practice closer to Green IT validation routines based on metering, calibration, and error reporting. It can also improve transparency and comparability and provide a more reliable basis for allocation decisions [36,40].

Critical research gap. The reviewed studies indicate a fragmented state of research: FLOP- or GPU-hour proxies are often reported without validation against metered baselines. Telemetry data is rarely linked to shared calibration protocols, and carbon-intensity traces are not yet systematically integrated into AI workload schedulers. A central gap concerns benchmarked approaches that combine workload profiling, metered baselines, and carbon-aware dispatch across cloud, edge, and on-premises infrastructures. Addressing this gap would strengthen the implementation relevance of Green AI research by connecting measurement, calibration, and workload-allocation decisions.

5.2. Operational pathways toward lifecycle-oriented Green AI

Despite these challenges, the reviewed literature suggests several research directions that can support more lifecycle-oriented Green AI. The synthesis points to three recurring research directions. One direction is to combine multi-level instrumentation — such as rack-level PDUs, processor-level performance counters, and local power meters — with AI-specific workload management. This can support telemetry-informed control, including DVFS and RAPL-based monitoring or power caps where supported by the hardware platform. Second, multi-object-

ive benchmarking and reporting protocols could integrate carbon intensity, cooling-related water indicators, and embodied hardware emissions, provided that data sources and assumptions are explicitly declared. Third, estimator-based approaches should be calibrated against metered reference data where available to improve measurement consistency across heterogeneous infrastructures.

Extend lifecycle footprint analyses beyond carbon-centric indicators. Future studies should broaden footprint analyses beyond training and explicitly declare which AI pipeline stages are included, such as data collection, preprocessing, deployment, inference, and model retirement [39,67,68]. Compared with reviews that focus primarily on software-level energy use or training-related emissions, this synthesis gives greater attention to embodied hardware-related impacts from production, transport, use, and end-of-life treatment [41], and to the need for clearer Scope 1–3 boundary declarations [17]. Multi-dimensional indicators should therefore complement carbon-centric accounts. Relevant examples include water withdrawals and consumption in cooling-intensive regions, Water Usage Effectiveness (WUE) in data centers, and material-scarcity indicators for critical raw materials [8,16]. This expansion can provide more informative analytical baselines for decisions on siting, procurement, and workload management.

Develop comprehensive monitoring tools. Monitoring tools that are both easy to use and practically integrable into AI development workflows remain limited. Future research should not be limited to GPU, CPU, and DRAM counters but should more systematically account for networking, storage, and cooling overheads, since these can materially affect data-center-level footprints and are often outside the scope of AI-specific counters [43]. Toolchains that connect resource metering with operational decision support remain underdeveloped. Examples include identifying energy-intensive data-loading steps, suggesting efficiency-relevant configuration changes, and supporting carbon-aware cloud-region scheduling. Such toolchains would make monitoring more directly usable for AI design and deployment decisions.

Toward multi-objective resource allocation for Green AI workloads. Much of the current literature, including several reviews, still emphasizes carbon- or energy-centered indicators. Future research should explicitly address multi-objective trade-off evaluations, for example, by combining carbon intensity and water-stress indicators in workload scheduling while accounting for embodied hardware impacts in longer-term allocation and procurement decisions. Such work could support allocation models that consider multiple Green AI objectives instead of optimizing a single environmental metric.

6. Discussion

Modern AI application systems increasingly combine heterogeneous workloads — e.g., spanning training, inference, and data management — under service-level and latency constraints. Such integration can affect peak-power requirements, “Scope 2” emissions, and hardware refresh cycles, thereby linking environmental impacts more directly to infrastructural design and operation [55]. Based on 103 reviewed articles, this review develops a lifecycle-oriented analytical synthesis of Green AI, including its definition, lifecycle structure, hardware contexts, and measurement dimensions. Earlier surveys have mapped broader environmental implications of AI [23], aligned AI with climate mitigation and policy concerns [32], or systematically characterized Green AI research topics, phases, artifacts, and tool availability [9]. In contrast, this analysis uses an LCA-oriented reference perspective to relate AI-specific lifecycle phases to environmental assessment concerns, including embodied and end-of-life impacts, while drawing on established concepts from “Green IT” and HPC. Relevant trade-offs therefore extend beyond carbon emissions to water use and material-related impacts. Measurement and scheduling are closely linked: carbon- or energy-aware scheduling depends on reliable estimates, and estimator-based assessments should be calibrated against metered reference data where available to improve consistency [40,53].

6.1. Theoretical and practical implications

Theoretically, Green AI should not be reduced to training-time efficiency gains [9,14,56]. Earlier surveys have mapped broader environmental implications of AI, aligned AI with climate mitigation and policy concerns, or characterized software-centric Green AI topics, artifacts, phases, and tool availability [9,23,32]. In contrast, this article contributes a review-derived, lifecycle-oriented analytical synthesis that relates AI lifecycle phases to LCA-oriented assessment stages without claiming to perform a full LCA. This perspective makes environmental burdens more visible that are often underrepresented in carbon-centric accounts, including Scope 3 emissions, embodied carbon, water use, and resource depletion [7,8,22,56].

Practically, the findings suggest directions for applying lifecycle-oriented Green AI in measurement, reporting, and workload-management contexts. Whereas Verdecchia et al. [9] provide a software-centric systematic review of Green AI topics, phases, artifacts, and tool availability, this synthesis additionally foregrounds how established Green IT and HPC concepts — such as DVFS, RAPL-based power limits, queue-level energy budgets, and checkpointing — can inform AI pipeline operation [40,56]. When adapted to AI contexts, such concepts can inform workload placement across heterogeneous hardware, power-aware configuration choices, and locality-aware scheduling that reduces avoidable data movement [17,23,40]. Prior reviews have addressed several of these elements, but the operational connection between Green IT/HPC concepts and lifecycle-oriented Green AI remains underdeveloped [9,56].

Beyond carbon intensity, operational allocation and longer-term infrastructure decisions should also consider water use and embodied hardware impacts [7,8,23,56]. Future multi-objective allocation approaches could extend workload scheduling by combining grid-intensity signals with facility-level water indicators and longer-term estimates of embodied hardware impacts [7,8,17,56]. A possible research direction is closed-loop scheduling, in which hardware counters, rack-level power data, and estimator outputs are compared to improve calibration and inform scheduler decisions in suitable workload-management systems [17,40,41]. Such approaches could help placement decisions consider regional grid intensity and cooling-related water constraints in addition to compute throughput [8,17]. Calibration deltas should be explicitly documented where estimator outputs are compared with metered reference data [40,41]. Where estimator outputs are used to inform scheduling, studies should report whether calibration deviations would affect the resulting placement or deferral decisions [17,40,41]. Such documentation supports reproducibility and cross-study comparison [40]. Finally, the analysis highlights that Green AI is shaped by material dependencies and infrastructural path dependencies [7,23,56]. AI hardware depends on semiconductor supply chains whose fabrication stages are energy- and resource-intensive [7, 56]. Short hardware refresh cycles can increase embodied impacts and waste unless operational efficiency gains offset the additional production impacts [7,23,56]. These structural factors can offset software-level efficiency gains and tie AI development more closely to material and energy constraints [7,23,55,56]. By including end-of-life, resource recovery, and circularity in the lifecycle perspective, the synthesis makes such structural dependencies more explicit and identifies policy-relevant intervention points, for example, around hardware lifetime extension, renewable procurement, and embodied-impact disclosure [7, 56].

6.2. Limitations

Despite its comprehensive scope, this review has limitations. First, the database-driven search, while systematic, may have underrepresented relevant contributions outside computer science venues, particularly in engineering, sustainability science, and political economy.

Table 2
Summary of principal findings per research question.

Research question	Principal findings
(1) How should <i>Green AI</i> be defined and delimited relative to adjacent concepts, and which recurring principles are discussed in the literature in relation to its lifecycle-wide application?	• Concise definition of Green AI focused on cumulative environmental impacts across declared system and lifecycle boundaries while meeting specified performance requirements • Clear delimitation of Green AI from broader Sustainable AI concepts • Separation of conceptual definition and operational interpretation, including declared assessment boundaries, performance requirements, trade-offs, and uncertainty.
(2) Which phases and subphases structure the <i>Green AI lifecycle</i> , and how can they be analytically related to the stages of Life Cycle Assessment (LCA)?	• Five-phase lifecycle structure with 33 review-derived analytical subphases • Analytical relation of lifecycle activities to the four LCA stages without implying one-to-one equivalence • Explicit coverage of provisioning, development, deployment/use, maintenance, and end-of-life/circularity across the lifecycle.
(3) Which hardware components and system architectures are associated with improved performance-per-watt and potentially reduced embodied environmental impacts across the lifecycle?	• Class-based hardware and infrastructure mapping across edge, workstation, on-premises, and cloud contexts • Hardware-related results depend on contextual variables such as region, PUE, timing, workload profile, and deployment setting. • Shift from device-level examples to context-dependent architectural trade-offs and performance-per-watt considerations.
(4) How can environmental impacts (e.g., energy, CO ₂ e emissions, water use, and embodied/material footprints) be <i>measured and reported</i> consistently across the lifecycle, including calibration of indirect estimators with metered or telemetry-based reference data?	• Hybrid measurement logic that relates software estimators to telemetry and metered reference data across different measurement levels • Provider or corporate sustainability dashboards are useful but ecosystem-bound and therefore limited in cross-system comparability. • Water- and lifecycle-related indicators remain weakly operationalized and require clearer system boundaries, proxy validation, and more standardized reporting.
(5) Which open <i>challenges and research priorities</i> , such as benchmarks, reporting schemas, uncertainty handling, circularity, and governance-related conditions, are needed to advance lifecycle integration of Green AI?	• Research priorities include benchmark suites, reporting schemas with calibration information, uncertainty disclosure, and stronger integration of carbon, water, and embodied/material indicators. • Analytically identified operational directions include telemetry-informed calibration, carbon-aware scheduling, and selected HPC-inspired resource-management concepts such as DVFS, RAPL-based caps, energy-aware placement, and checkpointing for AI workflows. • Open challenge: validating WUE/cooling-water estimates where facility-level measurements are available and linking calibration, allocation, and reporting across heterogeneous infrastructures.

Second, the integration of LCA with AI lifecycle phases remains conceptual and is not empirically validated. It serves as an analytical synthesis of literature-derived patterns, not as a validated operational framework. Practical application would require empirical assessment, such as case studies or expert evaluation. Third, although the review-derived classification spans 103 studies, continuous revision is necessary as standards, tools, and policies evolve. Beyond technical indicators, future research could examine how disclosure requirements, renewable-energy siting, and producer-responsibility regimes shape the distribution of environmental burdens. Future work should also examine regions involved in material extraction, hardware manufacturing, and e-waste processing, where AI-related infrastructure may contribute to unevenly distributed environmental burdens [8,16,55].

Overall, the review links lifecycle mapping with selected Green IT and HPC concepts as a structured synthesis of existing approaches. It outlines how phase-specific measurement considerations, calibration of estimator tools, and hardware-related accounting practices can be analytically linked within a lifecycle perspective, without implying a prescriptive or universally validated methodology. The lifecycle structure and analytical distinctions are synthesis-driven structuring devices, not validated standards, governance mechanisms, or universally applicable process models. They require further empirical validation.

7. Conclusion

This article consolidates key foundations of *Green AI* and develops a lifecycle-oriented analytical synthesis of the field. A lifecycle-oriented

perspective is combined with a class-based analytical view of hardware deployment contexts and a differentiated measurement logic that relates indirect software estimators to metered or telemetry-based reference data. The synthesis identifies operational directions — such as carbon-aware scheduling, the transfer of selected Green IT/HPC concepts, and calibration-oriented measurement practices — that can inform more transparent Green AI evaluation beyond training-only efficiency gains.

Practice and policy implications. As a practical implication of the synthesis, future AI studies should report environmental indicators alongside standard performance results. They should also disclose key assumptions in a standardized manner. This includes region-specific carbon intensity and temporal granularity. It also includes Power Usage Effectiveness (PUE). For deployment and scheduling studies, relevant elements include deferral windows, SLA constraints, migration and data-locality policies, telemetry sources, calibration error, and weighting assumptions across carbon, water, and embodied impacts. Reporting negative results and releasing datasets and code for Green AI experiments could reduce redundant computation and support methodological convergence [23,32].

Contributions. This article provides a structured and lifecycle-oriented analytical contribution to Green AI. The contribution consists of four elements. First, it articulates a synthesized conceptual definition of Green AI and a proposed review-based analytical completion definition of Green AI. Second, it synthesizes a *five-phase* lifecycle structure with 33 review-derived analytical subphases that analytically relate AI lifecycle activities to LCA stages. Third, it synthesizes

a *class-based* hardware perspective that relates deployment contexts, infrastructural conditions, and characteristic trade-offs. Fourth, it systematizes measurement methods across indirect estimators and direct approaches. These approaches span different telemetry levels, including PMCs, RAPL-based energy or power estimates, and rack-level PDU measurements. The synthesis also highlights limitations of water-use and embodied-emission proxies. Building on these contributions, the article identifies analytically grounded directions for lifecycle-oriented Green AI research and practice.

Research answers. To avoid dispersing the findings across the preceding sections, Table 2 summarizes the principal answers to RQ1–RQ5 and links them to the corresponding synthesized outcomes and contributions.

For RQ1 (“How should *Green AI* be defined and delimited relative to adjacent concepts, and which recurring principles are discussed in the literature in relation to its lifecycle-wide application?”), the review synthesizes a Green AI definition focused on reducing cumulative environmental impacts across declared system and lifecycle boundaries while maintaining specified performance requirements. It distinguishes Green AI from broader concepts such as Sustainable AI and proposes a review-based analytical completion definition of Green AI that makes assessment-relevant elements explicit, including reference conditions, trade-offs, and uncertainty.

For RQ2 (“Which phases and subphases structure the *Green AI lifecycle*, and how can they be analytically related to the stages of Life Cycle Assessment (LCA)?”), the synthesis produced a review-derived lifecycle structure with five phases and 33 subphases. These phases and subphases were analytically related to the four LCA stages. The five phases are “Green hardware selection and infrastructure design”, “Green AI development”, “Low-footprint AI task realization”, “Circular AI maintenance and green administration”, and “Green end-of-life and AI circularity”.

For RQ3 (“Which hardware components and system architectures are associated with improved performance-per-watt and potentially reduced embodied environmental impacts across the lifecycle?”), the review synthesizes a class-based hardware perspective spanning edge, workstation, on-premises, and cloud deployment contexts. Rather than prescribing devices, it relates hardware choices to contextual variables such as region, PUE, timing, workload profile, and deployment setting, thereby supporting context-dependent comparison of architectural trade-offs.

For RQ4 (“How can environmental impacts (e.g., energy, CO₂e emissions, water use, and embodied/material footprints) be *measured and reported* more consistently across the lifecycle, including calibration of indirect estimators with metered or telemetry-based reference data?”), the review synthesizes a hybrid measurement logic in which estimator-based approaches can be calibrated against metered or telemetry-based reference data, including RAPL readings, PMCs, rack-level PDUs, and smart plugs, depending on the measurement boundary. Provider dashboards, while useful, remain ecosystem-bound and water- and lifecycle-related indicators are still weakly operationalized, limiting comparability across heterogeneous infrastructures.

RQ5 (“Which open *challenges and research priorities*, such as benchmarks, reporting schemas, uncertainty handling, circularity, and governance-related conditions, are needed to advance lifecycle integration of Green AI?”) highlights that lifecycle integration remains constrained by fragmented reporting, lack of widely adopted calibration standards, limited benchmark availability, and weak integration of carbon, water, and embodied impacts. Based on these gaps, future work should strengthen telemetry-informed calibration, standardized reporting schemas, uncertainty disclosure, multi-objective workload allocation, and facility-level validation of water-use estimates to improve reproducibility and cross-infrastructure comparability.

Outlook. Future work should therefore explore three directions. First, hybrid measurement models should include declared calibration against metered reference data. Second, benchmark suites should

combine representative AI workloads with temporally resolved carbon intensity, regional water-use indicators, and embodied-carbon assumptions. Third, scheduling approaches should make SLA-related trade-offs more explicit by reporting deferral windows, operational constraints, and environmental effects where available. Embedding Green AI in such operational practices could support more reproducible cross-study comparisons. It could also shift evaluation beyond narrow efficiency gains toward lifecycle-oriented Green AI assessment [9,32,55].

CRediT authorship contribution statement

Marcel Rojahn: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Marcus Grum:** Writing – review & editing, Validation, Supervision, Resources, Project administration, Funding acquisition.

Declaration of Generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used ChatGPT (OpenAI) for language refinement and clarity checks and LanguageTool Premium (LanguageTool GmbH) for spelling and grammar support. After using these tools, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.infsof.2026.108186>.

Data availability

Supplementary tables, coding summaries, and analytical materials supporting this systematic literature review are provided in the [Online Appendix](#). The full extraction sheets are available from the corresponding author upon reasonable request. No further new primary datasets were generated. The analysis is based on the literature corpus reported in the article and on review-derived extraction and coding materials.

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